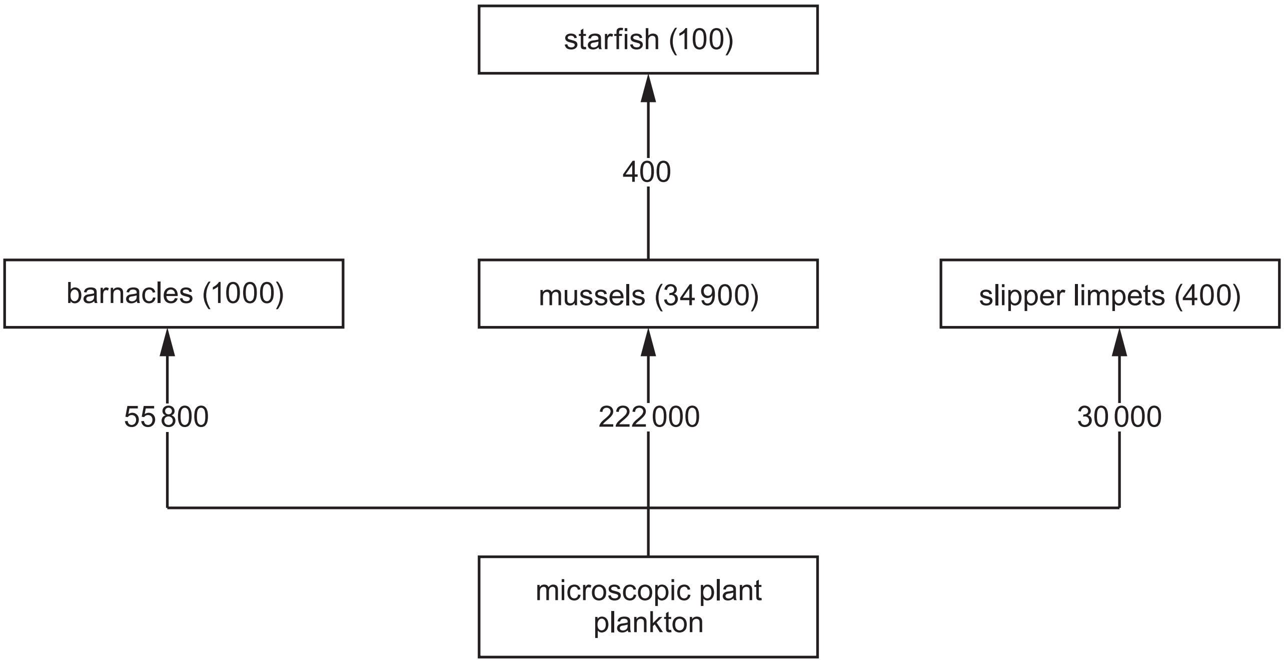
Examiner only

7. The flow diagram shows the transfer of energy between organisms on a rocky shore. Numbers on the arrows show the energy available to the organisms in kJ per m^2 per year. Numbers in brackets show the energy that becomes part of the biomass of the organisms in kJ per m^2 per year.



The energy efficiency of an organism is a measure of how much of the energy available to the organism becomes part of its biomass. It is calculated by using the following equation:

$$\% \text{ energy efficiency} = \frac{\text{energy that becomes part of biomass}}{\text{energy available}} \times 100$$

(a) (i) Calculate the energy efficiency of the starfish. [2]

energy efficiency = ... %

(ii) Calculate the percentage of the energy entering the mussels that enters the starfish to **three** significant figures. Show your working. [3]

percentage of the energy entering the mussels = ... %

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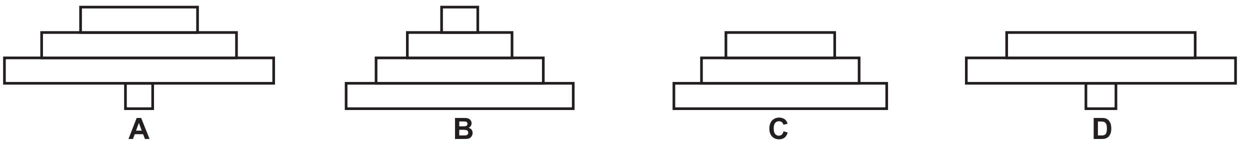
(b) The diagrams show pyramids of numbers for four ecosystems.



Which **one** of the pyramids of numbers represents the flow diagram of this rocky sea shore? State a reason for your answer. [2]

....

....

7

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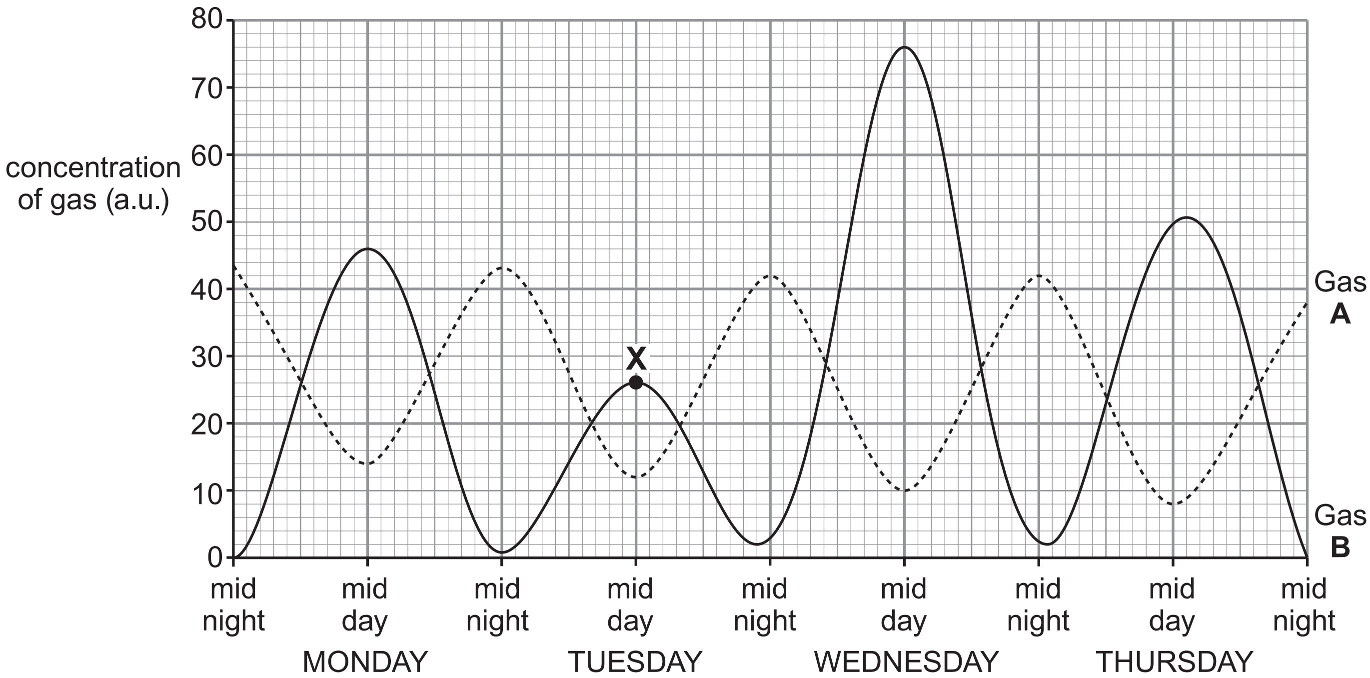
8. The graphs below show changes in the concentration of two gases, **A** and **B**, in a rock pool in St Brides Bay, Pembrokeshire over a 4-day period. The rock pool has a high biomass of plants.



(a) (i) Identify gases **A** and **B**. [2]

Gas **A** ....

Gas **B** ....

(ii) Explain the results at midday for gas: [4]

A; ....

....

....

B. ....

....

....

(iii) Suggest a reason for the lowest peak at point **X**. [1]

....

Examiner only

(iv) Calculate the mean maximum concentration of gas **B** over 4 days. [2]

mean maximum concentration of gas **B** = ... a.u.

(b) Name the limiting factor that can be deduced from the data in the graph. [1]

....

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